contribute, including developers outside the core group. External contributions accelerate feature releases and reduce vendor lock-in, allowing compatibility with any hardware and chip vendor. SONiC addresses growth, agility, speed, and dependency barriers, enabling a flexible and innovative networking environment.

Q. What challenges do open source networking solutions face in expanding their market share, especially when traditional networking software is still prevalent?

A. The first challenge is building confidence that open source software is reliable and readily available. SONiC, for instance, is proving to be more dependable than traditional solutions. This process is similar to what the Linux operating system went through in the compute space, but because of the Linux experience I believe the SONiC adoption process will be faster. The second challenge lies in transitioning from legacy networks, which involves adopting new hardware and software and retaining personnel.

While this shift presents its own difficulties, confidence is steadily growing due to the successful adoption of SONiC by major players like Microsoft, Dell, and Alibaba. Over time, I believe these challenges will diminish as trust in open source solutions continues to strengthen.

Q. How does this open source software drive innovation?

A. SONiC, as part of the Linux Foundation, is a product of efforts like the Open Compute Project (OCP), which was instrumental in its development. Innovation in areas such as artificial intelligence and security is advancing rapidly. This year, significant improvements were made in security and new features were introduced, including support for physical servers and pizza boxes. Previously, SONiC was deployed primarily on T1 and TOR configurations (pizza boxes) connected to physical servers. Now, it is moving towards modular chassis designs. Once this deployment is fully operational, enterprises and data centres, regardless of scale, can implement SONiC with any chip vendor.

Q. How scalable is the solution?

A. SONiC is highly scalable. It is successfully deployed in both enterprise settings and large scale data centres. Two major players with some of the largest networks in the industry, Microsoft and Alibaba, are using SONiC extensively.

For example, Alibaba's data centres grow very rapidly every month, and SONiC operates flawlessly at that scale. Its hardware-agnostic design ensures compatibility and seamless operation regardless of the underlying hardware.

Q. How does SONiC, as an open source networking solution, ensure security?

A. Security is paramount, especially for open source solutions. Every feature in SONiC is hosted on GitHub, where anyone can access and download it. However, before any code is approved for production, it undergoes a rigorous pull request (PR) review process, ensuring compliance with strict security criteria.

Beyond software security, we have implemented ring-fencing mechanisms with supplementary tools and connections to the chipset's software panel. Hardware vendors also contribute additional security measures. These dual layers—software and hardware—make SONiC a highly secure open source platform.

Q. How sustainable is an open source model in ensuring continuous development and innovation in networking solutions?

A. Open source solutions benefit organisations across all scales, from small enterprises to large corporations, making it very sustainable.

For instance, companies like Aviz Networks and BE Networks (formerly known as BeyondEdge) provide tools and services to deploy open source networks even on a small scale. For medium-sized setups that aim to avoid high costs but still require reliable, supportable solutions, SONiC offers comprehensive base package features with rapid release cycles to meet those needs.

Q. Could you elaborate on the significance of the community?

A. The community is extremely important. It is the community-driven development model that makes open source solutions like SONiC highly sustainable and reliable. Currently, we have around 16 major companies involved, including Google, Microsoft, Alibaba, Dell, Broadcom and others. NVIDIA is also a part of this network, both as a consumer and a supporter of SONiC.

The community is vital because feature development and new software releases rely on its contributions to meet market expectations. We have thousands of active developers contributing to coding, and feature releases occur frequently—about a hundred every two years. For instance, this year alone, we launched over ten new features. The community's consistent support and contributions are what enable SONiC to thrive and evolve.

Q. How do you encourage new contributors to join and collaborate on this project?

A. The beauty of open source is its universality. For example, if I develop a specific feature, such as IPv6 support or security-related functionalities, any consumer—irrespective of their hardware—can utilise it without waiting for their hardware vendor to implement the same feature. Feedback is important, whether it is about missing features, potential improvements, or areas where we are lacking. I am part of the SONiC Outreach Community, which serves as a platform for gathering this feedback